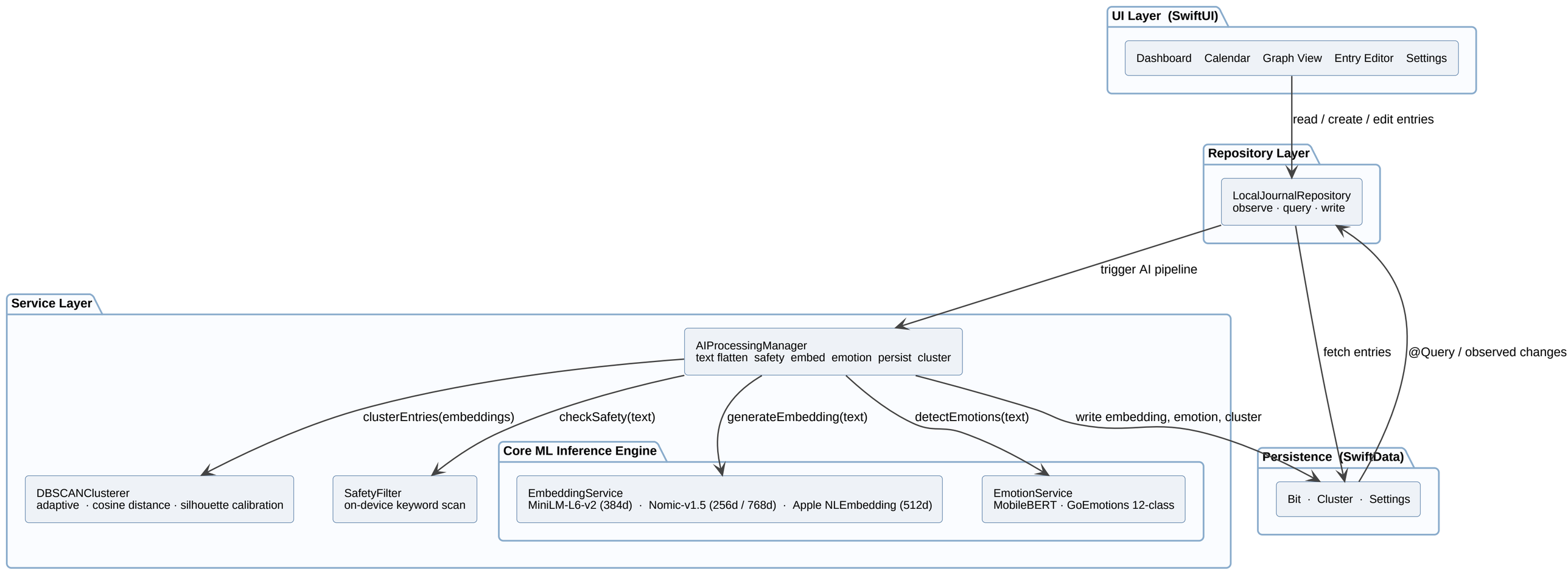
UI Layer (SwiftUI)
Dashboard
Calendar
Graph View
Entry Editor
Settings
read / create / edit entries
Repository Layer
LocalJournalRepository
observe · query · write
trigger AI pipeline
fetch entries
@Query / observed changes
Service Layer
AIProcessingManager
text flatten safety embed emotion persist cluster
clusterEntries(embeddings)
checkSafety(text)
generateEmbedding(text)
detectEmotions(text)
write embedding, emotion, cluster
DBSCANClusterer
adaptive · cosine distance · silhouette calibration
SafetyFilter
on-device keyword scan
Core ML Inference Engine
EmbeddingService
MiniLM-L6-v2 (384d) · Nomic-v1.5 (256d / 768d) · Apple NLEmbedding (512d)
EmotionService
MobileBERT · GoEmotions 12-class
Persistence (SwiftData)
Bit · Cluster · Settings